Team Kestrel

Business Case Document

One place for every path after A/L results

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1. The problem

After A/L results, a student has a few weeks to choose between many paths, and the rules for each sit in a different place.

In 2025, 281,810 students sat the A/L and 176,527 qualified for university [1], and the state system offered 42,937 places [2]. So about 133,600 qualified students, roughly three in four, will not get a state place and must find another path. Those aiming for a state university must pick and rank up to 125 courses within a three-week window, under rules in a 200+ page UGC handbook that changes every year [2].

The paths run on their own calendars:

Path	Where the rules live	When it moves
State university	UGC handbook, yearly cut-offs, UGC notices	Fixed 3-week window, then later rounds of about one week
Private or non-state degree	Each institute's own website and fairs	Each institute sets its own intake, fees and entry rules
Gazetted job exam (e.g. Customs)	Weekly Government Gazette	A notice can appear any week
Vocational course	Separate institutes and notices	Rolling intakes
Study abroad	Scholarship and university sites	Varies by country
Others (professional bodies, diplomas, etc.)	Each body's own website	Own intakes and dates

With no single source, students decide from tuition teachers, seniors and Facebook groups. The cost of a wrong move is high:

- **A badly ordered list.** Selection follows the order of preference, so a Safe course listed first blocks the courses the student wanted more.
- **An ineligible choice.** Some courses need a specific subject grade or an aptitude test that the student never checked.
- **A missed date.** Missing the application window loses a year; missing a later round loses the place.
- **An unknown path.** A job exam or scholarship the student never heard of.

The National Education Commission found most students and parents unaware of alternative education and career options [3], and school guidance "more substantial on paper than in reality", especially in rural areas.

Evidence from students

Six conversations with people at many points on the path showed one pattern: nobody knew where to look.

We spoke to a current A/L student, two private-university undergraduates, a student on an alternate path, a foreign-scholarship holder and a state-university undergraduate (E1 to E6), between 23 and 29 September 2026. All six conversations are recorded [4].

What we heard	People	In their words
Didn't know the other paths, or found them by luck	4 of 6	"Basically it's luck." (E5, on a foreign scholarship)
Couldn't follow the application or the handbook	3 of 6	"No one can really read the campus book. I got my info from YouTube." (E2)
No single source; relied on parents, friends and YouTube	3 of 6	"I had no proper way to find out, like about the Gazette." (E3)
One gap changed their path	2 of 6	"I missed one course… life altering situation." (E4)
Wanted to know what a degree actually teaches	2 of 6	"If you could show the curriculum, that would be good." (E6)

Two requests shaped the product directly. E5 asked us to "show all the options he can take after the Z-score is released, and the steps". E6 warned that student groups must "first check that the seniors and admins are legit", so ZedPath verifies every senior before they can post.

We also tested three existing cut-off tools hands-on with one real profile (Physical Science, Kurunegala, Z 1.4821). All compared the Z-score with cut-offs. None checked subject grades, the preference order or deadlines. For example, Computer Science at Ruhuna needs a C in Combined Maths or Physics, yet each tool listed it as eligible without asking for grades.

2. Proposed solution: ZedPath

ZedPath turns the handbook, the cut-off tables and the Gazette into checks a student can run on their own results in under a minute.

ZedPath is a free, trilingual web that also comes as a native app. It rates every state course Safe, Likely or Reach from five years of district cut-offs and lists the other paths. It checks the student's preference order and tracks deadlines with reminders. Ask ZedPath, its AI assistant, answers in Sinhala, Tamil or English only from the official sources it retrieves, and every answer shows the page it came from so the student can check it.

Feature	What the student gets	Problem it removes
Enter results once	Stream, district, Z-score and subject grades, entered one time and reused everywhere	Re-checking the same facts on every site
Every path	All 255 state course codes (Uni-Codes) rated Safe, Likely or Reach from five years of district cut-offs [2][6], beside private degrees, job exams, vocational courses, study abroad, etc.	Not knowing the other paths
Course pages	Entry rules checked against the student's own grades, aptitude-test flags, the cut-off history and the official syllabus	Ineligible choices; "what does this degree teach?"
My list	The UGC preference list (up to 125 courses), with warnings when the order blocks a course the student wants more	A badly ordered list
Compare	Up to three courses side by side: chance, length, first-year subjects, extra steps	Choosing between degrees blind
Ask ZedPath	Questions answered in Sinhala, Tamil or English from the UGC handbook and official pages, each answer citing its page and linking to the official source to check; off-topic questions refused	Hearsay and unreadable handbooks
Gazette alerts	Every weekly Gazette read; a new job exam the student qualifies for becomes an alert with its closing date	Missing job exams
Journey and reminders	Every step from results to registration, marked confirmed or estimated, with reminders as phone notifications	Missed windows and later rounds
Course groups	A group per target course, verified seniors marked with a badge, report and block on every message	Unverified Facebook advice
Three languages	The whole interface in Sinhala, Tamil and English; native-speaker review to follow	English-only tools

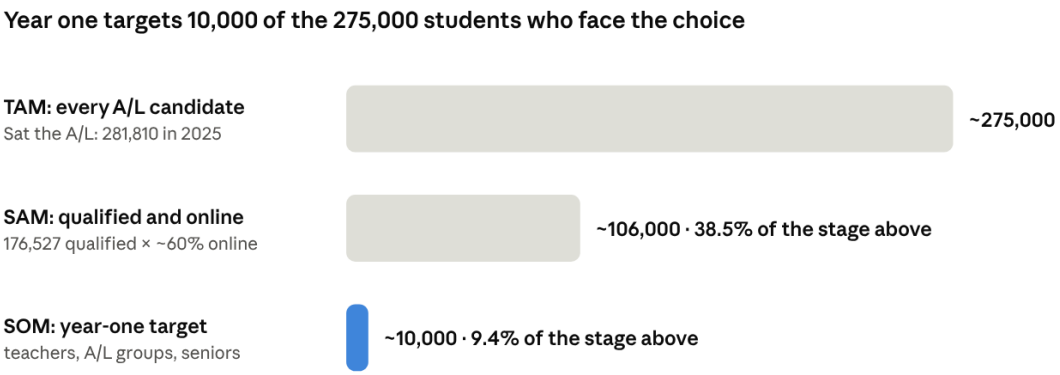
Feature	What the student gets	Problem it removes
Account	Required at the start: full name, email, Sri Lankan mobile and date of birth, with recorded consent; password hashed on the phone, delete everything in one tap	Losing a plan when the phone changes

ZedPath runs in any phone browser, installs like an app, and ships as a native Android app with its own notifications.

Trust rules ZedPath keeps. Every number links to its official source. ZedPath never says "you will get in"; it rates chances and shows the cut-off behind each rating. There is no paid placement: no one can pay for a higher position. Every detail stored is listed at sign-up with its purpose and needs the student's consent, and a student can delete everything at any time.

3. Target audience

About 275,000 students face this choice every year, and a new cohort arrives with every set of results, so the market renews itself.



Department of Examinations, A/L results 2023 and 2025 · DataReportal, Digital 2026: Sri Lanka · year-one target is our estimate

The user is a student between A/L results and their next step, on a phone, often first in their family to apply. Parents, tuition teachers and school guidance teachers are not users we sell to, but they are how students find ZedPath.

How we reached each number.

Number	How we got it	Source
About 275,000 a year	Average of A/L candidates in 2023 (269,613) and 2025 (281,810)	[1]
176,527 qualified	Candidates eligible for university admission, 2025	[1]
42,937 state places	42,037 places across the 255 Uni-Codes plus 900 additional Arts places, 2025/26; ZedPath's own extraction of the handbook adds up to the same total	[2]
About 106,000 reachable	176,527 qualified × about 60% internet use	[1], [5]
10,000 in year one	Sum of the channel estimates below; our assumptions, to be tested in the first cycle	Team estimate

Year-one channel	Basis (our estimate)	Students
A/L Facebook groups	5 groups × ~20,000 members × 5%	5,000
Tuition teachers	20 teachers × ~150 students	3,000
A/L Telegram channels	2 channels × ~65,000 subscribers × 1%	1,300
Seniors and friends sharing	Word of mouth	700
Total		**10,000**

4. Value proposition

For students: every path they can reach, checked against their own results, with the source behind every number, free.

For sponsors: a visible education CSR story ("helped 10,000 students plan their next step") and their name in front of students just as they open their first bank account or choose a mobile plan for university. Their name can appear anywhere in the app.

Against the alternatives

Students already patch the answer together from eight sources; none combines their own results, every reachable path and dated deadlines.

Alternative	Why students use it	Where it breaks
Seniors and relatives	Trusted and free	Advice from one year and one course
WhatsApp and Facebook groups	Fast answers from peers	Unverified and often contradictory
Tuition-class seminars	A teacher they trust	One list for the whole class, no personal check
UGC handbook and website	The official source	200+ pages, no personal answer, no other routes
Cut-off tools (findmydegree.lk, Unicompass, induwara.lk)	Quick cut-off lookup	No grade, order or deadline checks (our hands-on test)
ThuSh LK (site, Telegram, YouTube)	Degree details, comparisons, cut-off trends	A reference guide: doesn't read your results, check your order or track Gazette exams
Gazette and department notices	Official source for job exams	Hard to find, no alerts, easy to miss
Institute counsellors and fairs	Cover private degrees	Each promotes its own programmes

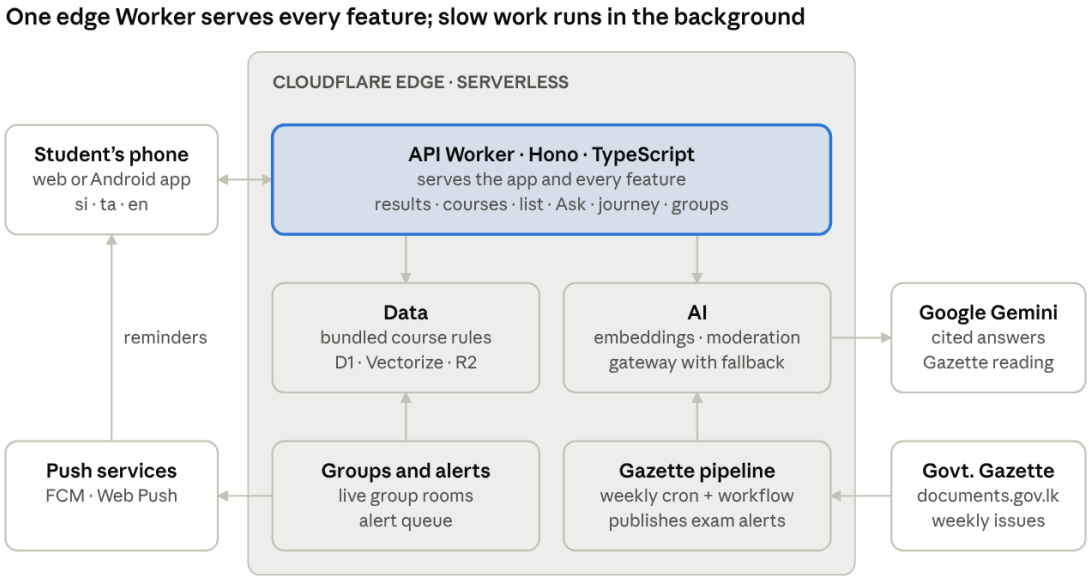
ZedPath's edge is the combination, and four things no tested tool does:

1. **Checks the student, not just the cut-off.** Subject-grade rules and aptitude tests are applied before a course is called reachable.
2. **Checks the order.** The preference list is tested against how selection actually works.
3. **Covers personalized paths outside the UGC.** Using the details each student gives when registering (stream, district, results, age and achievements), ZedPath checks which private degrees, job exams, vocational courses and study-abroad options they can apply for, and its AI assistant answers questions about them for that student.
4. **Tracks time.** Deadlines, later rounds and new Gazette exams become dated reminders.

5. Implementation

How it is built

ZedPath is built and running on a serverless stack that costs almost nothing when idle and scales on results day without anyone touching it.



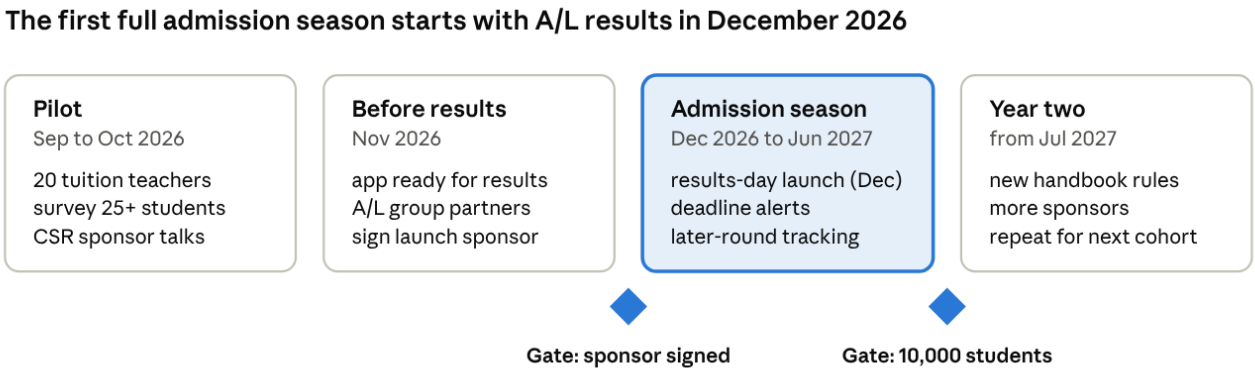
ZedPath architecture · simplified

- **One codebase, TypeScript throughout.** A React web app and a Hono API run in one Cloudflare Worker at the edge, close to students in Sri Lanka. The Android app is a native shell around the same app, so every web update reaches it without a new install.
- **Fast on mobile data.** The course rules and five years of cut-offs are compiled into the app's server bundle, so checking a student's results needs no database reads.
- **Official sources only.** The UGC handbook 2025/26, UGC cut-off tables 2021/22 to 2025/26, official route pages, university curriculum pages and the Government Gazette. Every fact shown links to its source.
- **Rules built to be checked.** Two independent extraction methods and two AI models must agree before a rule is published, and automated tests run against hand-verified cases before every release.
- **AI that cites.** Ask ZedPath searches multilingual embeddings of the official pages, answers through an AI gateway with a model fallback chain, and keeps only answers that cite a retrieved page.

Go-to-market and roadmap

We reach students where they already decide (tuition classes, A/L groups and seniors) and time every push to the admission calendar.

- **Teachers first.** A tuition teacher who shares ZedPath with a class reaches about 150 students at once, and teachers gain a tool that checks each student individually.
- **Results day is the launch.** Interest peaks when results are released; ZedPath must already hold the new handbook rules by then.
- **Reminders bring students back.** Saved lists, deadline alerts and Gazette exam alerts give a reason to return through the four-month season.
- **Seniors spread it.** Verified seniors in course groups are both a feature and a channel: each one is a trusted voice inside the next cohort.



Each cycle repeats with a new cohort, so the work of year one (rules pipeline, partners, channels) is reused every year.

Business model and costs

ZedPath is funded from several small income streams, so no single partner decides its future. The core stays free for every student, and no stream can buy a higher place in a student's results.

Income stream	What the payer gets	Price	Year-one estimate (LKR)
ZedPath Plus (optional, for students)	More Ask ZedPath questions a day, a printable application pack (ordered list, checklist, deadlines) and SMS or WhatsApp reminders. Results, every path, chances, Alerts, Journey and Groups stay free	LKR 490 per admission season	98,000 (2% of 10,000 students)
CSR and grants (banks, mobile operators, education funds)	Named support for one admission cycle, with their name shown across the app	One grant	100,000 (1 grant)

Year-one costs are small, because the app runs on serverless infrastructure and the founder and volunteers cover the language review and data checking. Figures use about LKR 300 per USD.

Cost	Year one (LKR)
Cloudflare Workers paid plan, USD 5 a month [8] (ZedPath runs on the free plan today)	18,000
Usage above the plan's included amounts at 10,000 students: requests stay inside the 10 million a month included; search and group moderation add about USD 10 in the season [8]	3,000
AI answers: about 100,000 questions on Gemini 2.5 Flash, about 3,000 input and 250 output tokens each (USD 0.30 and 2.50 per million tokens) [10]	46,000
Domain name (about USD 12 a year)	3,600
Google Play developer account, one time (USD 25)	7,500
Native-speaker review of the Sinhala and Tamil interface	0 (volunteers)
Data checker for the admission season (the founder or a volunteer)	0 (founder or volunteer)
Total	**78,100**

That is about LKR 8 per student a year, of which AI answers are about LKR 4.60. Plus at 2% of students brings LKR 98,000, which covers year one's cost (LKR 78,100) with a margin. One CSR grant (LKR 100,000) funds the next cycle's growth, for LKR 198,000 in all.

Risks and mitigations

The biggest risk is a wrong rule shown with confidence; every safeguard below exists to stop that.

Risk	Effect	Mitigation
A rule or cut-off is extracted wrongly	A student drops a course they could have got	Two independent extraction methods must agree; every rule is tested against hand-checked cases and links to its source page
The AI answers beyond its sources	Confident hearsay, the thing ZedPath replaces	Answers only from retrieved official pages; uncited claims are removed; off-topic questions are refused
Official formats change (handbook layout, Gazette site)	The yearly update or weekly reading breaks	Each run checks its output against the last; a failed check holds the update and alerts us instead of publishing
Students read "Likely" as a promise	False confidence, then disappointment	Never "you will get in"; the cut-off and year behind each rating are always shown
Unverified or harmful posts in groups	Misinformation, harassment	Seniors verified before posting; automatic moderation; report and block on every message
Personal data exposure	Loss of trust	Only four contact fields, each with a stated purpose and recorded consent (Personal Data Protection Act No. 9 of 2022 [9]); contact details kept apart from results; passwords hashed on the phone; one-tap delete; rate limits on sign-in
Revenue pressure to favour paying sponsors	Breaks the core promise	No paid placement is a written rule; sponsors and Plus subscribers never change a ranking or an answer
Usage spikes on results day	Slow or failed pages when it matters most	Serverless edge hosting scales automatically; course data is bundled with the app, so a results check needs no database reads

References

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4. Team Kestrel, *Gate 1: Problem and Proof*, IntelliCon '26 Buildathon: six recorded interviews (E1 to E6, 23 to 29 September 2026) and a hands-on test of three cut-off tools (27 September 2026).
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6. University Grants Commission, *Minimum Z-scores for selection to courses of study, by district*, academic years 2021/22 to 2025/26.
7. Government of Sri Lanka, *Gazette of the Democratic Socialist Republic of Sri Lanka*, Part I Section (IIA), issues of 28 August, 11 September and 18 September 2026 (examination and intake notices shown in ZedPath Alerts).
8. Cloudflare, *Workers pricing* and *Workers AI pricing*, developers.cloudflare.com, checked 11 October 2026 (Workers paid plan: USD 5 a month with 10 million requests included; Vectorize: 50 million queried dimensions a month included, then USD 0.01 per million; Workers AI: 10,000 neurons a day free, then USD 0.011 per 1,000).
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10. Google, *Gemini API pricing* (Gemini 2.5 Flash: USD 0.30 per million input tokens, USD 2.50 per million output tokens), ai.google.dev, checked 11 October 2026.